

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			colorimetry	1			1		1
	(b)	(i)		Cu^{2+} (1) absorbance reaches 0.1 in the shortest time / curve has the steepest gradient (1) neutral answer – has the highest rate			2	2		
		(ii)		change in absorbance in first ten seconds = 0.39 (1) 0.039 s^{-1} (1) accept 0.038-0.040			2	2	1	
		(iii)		$1 \times 10^{-4} \text{ mol}$		1		1		
	(c)			Indicative content <u>Temperature</u> Boltzmann distribution has a curve to indicate a higher temperature (curve drawn with peak lower and to the right) increasing temperature increases the (kinetic) energy of the particles more particles have energy greater than or equal to the activation energy more successful collisions per second / increased chance of successful collisions <u>Catalyst</u> Boltzmann distribution showing lower activation energy for reaction with catalyst [line for $E_a(\text{cat})$ drawn to the left of uncatalysed E_a] catalyst provides alternative reaction pathway / provides a surface for particles to adsorb to this lowers the activation energy more particles have energy greater than or equal to the activation energy	6			6		

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				5-6 marks Detailed description of both factors <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The information included in the response is relevant to the argument.</i> 3-4 marks Detailed description of one factor OR incomplete description of both factors <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. Mainly relevant information is included in the response but there may be some minor errors or the inclusion of some information not relevant to the argument.</i> 1-2 marks Incomplete description of one factor OR correct Boltzmann distributions with no written explanation <i>There is a basic line of reasoning which is not coherent, supported by limited evidence and with very little structure. There may be significant errors or the inclusion of information not relevant to the argument.</i> 0 marks No attempt made or no response worthy of credit.						
				Question 9 total	7	1	4	12	1	1